

# The Cosmochrony Research Programme: A Structural Roadmap and Paper Inventory

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## Abstract

The Cosmochrony programme develops a pre-geometric framework in which physical structure — spacetime, quantum mechanics, gauge symmetry, and Standard Model observables — arises from a single primitive: the local structure of admissible non-injective transitions between observable states. The corpus now comprises more than 50 papers across three theory branches. Branch I (Foundation track) builds the axiomatic base: four axioms suffice to derive the Heisenberg group  $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$  and its Weil representation as theorems, without any background substrate or relaxation mechanism. Branch II (O-series) develops the spectral admissibility sub-programme, deriving and numerically validating the transfer chain  $c_\chi \rightarrow \delta_{\text{pair}} \rightarrow \beta^* \approx 0.126$  connecting the BI saturation constant to the charged-lepton mass hierarchy. Branch III (Q-series and companion papers) derives quantum mechanics, spacetime geometry, gauge structure, and further physical observables from Branch I axioms and Branch II spectral data. This paper provides a complete inventory, a logical dependency map, and a structured account of what is proved, structural, heuristic, or open. It is intended both as an entry point for external readers and as an internal navigation reference.

**Keywords.** Non-injective projection; admissibility; Born–Infeld constraint; spectral admissibility; Heisenberg group; Weil representation; emergent spacetime; lepton mass hierarchy; quantum mechanics; gauge structure; programme overview.

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# 1 Introduction and Purpose

The Cosmochrony programme was initiated with a white paper [13] positing that all physical structure emerges from a non-injective projection  $\Pi : \Omega \rightarrow \mathcal{O}$  from a pre-geometric substrate  $\chi$  onto observable space. The initial framework postulated a Born–Infeld (BI) saturation constraint, a relaxation mechanism, and an iterated projection cascade  $U_{t+1} = \Pi \circ \sigma(U_t)$  as the engine of dynamics.

Over the course of 2026, the programme has evolved in two parallel directions. On one side, the spectral admissibility sub-programme (O-series, papers O1–O28) has progressively built the computational machinery for extracting physical observables from the spectral structure of the substrate, culminating in the proof of the unconditional transfer chain in O24 [55]. On the other side, the Foundation paper [4] has refounded the entire framework from four axioms (A1–A4), dispensing with any explicit substrate  $\chi$  and relaxation mechanism: these emerge as consequences of the axiomatic structure rather than being postulated. The Q-series then builds quantum mechanics,  $SU(2)$  symmetry, and Lorentzian spacetime on top of the Foundation axioms and O-series spectral data.

The corpus now contains more than 50 papers with complex mutual dependencies. A reader entering the programme faces two difficulties: it is not obvious where to start, and the distinction between a theorem, a structural argument, and an open conjecture is not always transparent from individual paper titles. This roadmap addresses both difficulties.

**Conventions on result status.** Throughout this paper each result is classified as one of the following: **proved** (formally proved as a theorem within the stated assumptions); **structural** (derived from structural arguments that are internally consistent but not yet fully formalised); **heuristic** (physically motivated and numerically supported but not yet rigorously derived); **open** (explicitly conjectured or known to be unsettled). These classifications follow the original papers; this roadmap introduces no new claims.

## 2 Conceptual Architecture

The programme is organised in three structural levels.

**Level 1 — Axiomatic primitive.** The primitive is the local structure of admissible non-injective transitions. No background spacetime, no quantum postulate, and no substrate field are introduced at this level. The four axioms are: local projective admissibility (A1), structural non-injectivity (A2), admissibility as non-premature selection (A3), and projection locking (A4). From A1–A2 alone, irreversibility and the arrow of time follow. From A3, the Heisenberg non-commutativity and the Weil representation emerge. From A4, the discreteness of quantum transitions follows. Papers at this level: ENI [28], Foundation [4], HeisenbergStructure, noscale [26].

**Level 2 — Spectral fibre structure.** The admissible fibre is proved to carry the Weil representation  $V_\rho$  of  $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$  for a prime  $q$  (Foundation Theorem 5.6; detailed proof in HeisenbergStructure [27]). The O-series computes the spectral properties of this fibre — in particular the capacity exponent  $\delta_{\text{pair}}$  and its transfer to the cascade exponent  $\beta^*$  governing the charged-lepton mass hierarchy. Papers at this level: SpectralAdmissibility [46], SpectralCapacity [47], SpectralGramRigidity [31], SpectralRelaxation [6], and O1–O29 ([37]–[48]).

**Level 3 — Physical observables.** Effective spacetime geometry, gauge structure, Standard Model charges and masses, gravitational waves, and cosmological observables are derived as consequences of Levels 1–2. No additional physical postulates are introduced beyond the continuum-limit hypotheses of Q5a–Q5b. Papers at this level: the Q-series (Q1 [18], Q2 [40], Q5a–Q6b), the

gravity track (Spectral2 [42], Gravity [21], Thermodynamics, Lorentz [11]), and the companion papers ([8, 10, 45, 12, 20, 54, 53]).

### 3 Branch I: Axiomatic Foundations

#### 3.1 The White Paper

The white paper [13] (“Cosmochrony: A Non-Injective Projection Theory of Emergent Physics,” v3.1) is the programme’s primary overview document. It introduces the substrate field  $\chi$ , the non-injective projection  $\Pi$ , the BI saturation constraint, the relaxation mechanism, and the qualitative derivation of spacetime and quantum structure. The spectral entropy functional  $S_\Pi[g] = \frac{1}{2} \log \det' A_g$  appears here as the structural origin of the Einstein equation.

*Role:* Programme overview; structural motivation; first formulation of the BI constraint and the relaxation cascade. *Evolution note:* The explicit substrate  $\chi$  and relaxation mechanism of the white paper are not postulated in the Foundation reformulation: they emerge from axioms A1–A4. The white paper remains the primary narrative reference; Branch I papers are the formal foundation.

#### 3.2 Non-Injectivity as a Structural Necessity of Genuine Emergence

The ENI paper [28] proves the following no-go theorem. Any genuinely emergent (non-isomorphic effective) physical description is necessarily non-injective and intrinsically compressive. The argument is purely structural: given a surjective  $\Pi : \Omega \rightarrow \mathcal{O}$  such that  $\mathcal{O}$  carries no independent microstate-resolving structure beyond  $\Pi$ , non-isomorphism of levels forces non-injectivity. A projection entropy  $S_\Pi \geq 0$  is defined, and the equivalence genuine emergence  $\Leftrightarrow \Pi$  non-injective  $\Leftrightarrow S_\Pi > 0$  is established (**proved**, Theorem 2.1 and Corollary 3.1).

*Role:* Foundational justification for structural non-injectivity (Axiom A2). Cited by Foundation, HeisenbergStructure [27], Q1 [18], Bell.

#### 3.3 Foundation: Admissible Non-Injective Transitions as the Primitive

The Foundation paper [4] establishes the minimal axiomatic basis of the programme. From the four axioms A1–A4 alone it derives: (i) irreversibility and the arrow of time (A1+A2); (ii) the proto-state as a physically real unresolved configuration retaining phase coherence (A3); (iii) a non-trivial commutator between conjugate generators, forced by A3 and the minimality of A1; (iv) the identification of the admissible fibre symmetry group as  $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$  for a prime  $q$ , together with BI parity (A1–A3); (v) the identification  $F_n \cong V_\rho$  (Weil representation) as a theorem (Theorem 5.6); (vi) the discrete character of quantum transitions from A4.

*Status:* **proved** for the algebraic identification of the fibre; **structural** for the continuum limit (deferred to Q5a–Q5b). *Role:* Axiomatic spine. Cited by all Q-series papers, HeisenbergStructure [27], noscale, and recent companion papers.

#### 3.4 Non-Factorisability and the Emergence of Heisenberg Structure

HeisenbergStructure [27] provides the detailed proof of Foundation Theorem 5.6, omitted from the Foundation paper for concision. The central structural lemma — *admissible non-factorisability* — is a direct formalisation of Axiom A3 and requires no input beyond A1–A3, BI parity, and classical group-classification results. As a corollary, the Heisenberg uncertainty principle is derived as a structural property of the non-resolved proto-state, not as an independent postulate (**proved**).

*Role:* Detailed proof of the algebraic core; cited by Q1 [18], Q2 [40].

### 3.5 No External Dimensional Scale at the Level of Admissibility

The noscale paper [26] establishes a structural rigidity result (Level 1): any deformation of the admissibility constraints that introduces a dimensional parameter independent of  $c_\chi$  breaks at least one of BFS-consistent spectral scaling, structural non-injectivity (A2/ENI), or bounded admissibility closure (A3–A4). Dimensional analysis within the framework is therefore fully constrained by  $c_\chi$  alone (**proved**).

*Role:* Closes the naturalness question for dimensional parameters.

## 4 Branch II: Spectral Admissibility Sub-Programme (O-Series)

The O-series targets the derivation of the charged-lepton mass hierarchy from the spectral structure of the admissible fibre. The central quantity is the power-law exponent  $\delta_{\text{pair}}$  of the pair-level capacity observable  $\sigma_{\text{pair}}(n) = \sigma_c(n) \cdot \sigma_{q-c}(n)$  along the BFS cascade. The main transfer chain is

$$c_\chi \longrightarrow \delta_{\text{pair}} \longrightarrow \beta^* \approx 0.126 \longrightarrow m_e : m_\mu : m_\tau. \quad (1)$$

This chain is proved unconditionally with respect to the fibre structure of  $\Pi$  (O24 [55], Corollary 4.2). The numerical validation of (1) is the subject of O25 [50].

### 4.1 Precursor Papers: SU(2) Admissibility Sub-Programme

Three papers precede the O-series proper and explore spectral admissibility on Cayley graphs of finite SU(2) subgroups.

**SpectralAdmissibility** [46] analyses the simplest case: the quaternion group  $Q_8$ . It derives the admissibility hierarchy  $A_n^{\text{max}} = c_\chi / \sqrt{\lambda_n}$  in the linearised regime, establishing the first analytically tractable instance of spectral admissibility (**proved**).

**SpectralCapacity** [47] introduces the capacity functional  $\mathcal{C}(G, S)$  and proves the binary-polyhedral maximality conjecture: the adjoint identity  $\hat{A}_{\text{adj}} = 2M - I$  (where  $M$  is the second-moment tensor of the axis distribution) shows that icosahedral isotropy forces  $\mu_{\text{adj}}(2I) = -\frac{1}{3}$ , while axis-plane confinement forces  $\mu_{\text{adj}}(\text{Dic}_n) \geq 0$  for all  $n \geq 3$ ; combined with a Plancherel uniform bound, this yields a strictly increasing capacity gap  $\Delta(\alpha) > 0$  for all  $\alpha > 0$  (**proved**).

**SpectralGramRigidity** [31] (v1.0.1) shows that the geometry of neutral generating sets on SU(2) is fully encoded in pairwise character values  $G_{st} = -\frac{1}{2}\chi_\rho(st)$  (**proved**).

**Three Particle Generations** [51] derives the three-generation structure from the ADE stratigraphic levels of binary Cayley graphs (**structural**).

### 4.2 SpectralRelaxation: Expander Graph Dynamics

The SpectralRelaxation paper [6] establishes  $\Lambda_{\text{proj}} \asymp \lambda_2$  for LPS expander families (**proved**), providing the entry point for the O-series dynamics.

### 4.3 O1–O8: LPS Phase — Cascade Exponent $\beta$

**O1** [37]: LPS saturation is a fixed-valence artefact; growing valence restores the pre-saturation window (**proved**). (Note: O2 was not written; the O3 derivation path was chosen instead.)

**O3** [35]: First quantitative SM connection. The phenomenological window  $\beta^* \in (0.09, 0.13)$  is derived from the charged-lepton mass ratios  $m_e : m_\mu : m_\tau$  (**structural**).

**O4** [49]: The structural upper bound  $\beta \leq 1$  follows from the BI flux constraint and the Cheeger isoperimetric inequality (**proved**).

**O5** [3]: Obstruction theorem. The  $\beta$  gap cannot be closed by any purely representation-theoretic or fixed-representation approach (**proved**).

**O6** [24]: Matrix-level confinement of the cascade exponent (**proved**).

**O7 [33]:** Reformulates the target as a capacity exponent  $\delta \in [7.4, 10.6]$  on three-step path fingerprints (**structural**).

**O8 [1]:** Establishes the geometric obstruction closing the LPS phase. Exponential shell growth  $|S_n| \sim p^n$  on LPS graphs compresses the pre-saturation window to  $O(\log q)$  BFS steps (**proved**).

#### 4.4 O9–O15: Heisenberg Transition

The geometric obstruction identified in O8 motivates replacing LPS graphs by Cayley graphs of  $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ .

**O9 [34]:** Structural transition. Polynomial ball growth  $|B_n| \sim n^4$  provides  $\Theta(q)$  usable BFS steps (**proved**). This is the key structural transition of the O-series.

**O10 [32]:** First large- $q$  computation ( $q = 101$ ). Confirms  $n^4$  growth; a TensorSketch representation obstruction is identified at large  $q$  (**proved**).

**O11 [56]:** Bidimensional Fourier-character proxy overcomes the TensorSketch obstruction (**structural**).

**O12 [15]:** Exact Weil-block fingerprints resolve the final obstruction. Establishes  $\hat{\delta}_{\text{exact}} \approx 4.3$ – $4.8$  for  $q \in \{29, 61, 101, 151, 211\}$  (**proved**).

**O13 [7]:**  $\hat{\delta}_{\text{exact}}(q)$  decreases monotonically from 4.80 to 3.59; the finite-size hypothesis is definitively refuted (**proved**).

**O14 [30]:** Corrects the  $\delta \mapsto \beta^*$  relation via block normalisation and central-phase contributions (**structural**).

**O15 [29]:** Block-level no-go. No reweighting of individual block data can raise  $\hat{\delta}_{\text{exact}}$  to  $\delta \in [7.4, 10.6]$  (Corollary 4.8, **proved**). This result forces the pair observable introduced in O16.

#### 4.5 O16–O24: Pair Observable and Unconditional Transfer Chain

**O16 [16]:** Identifies the misidentification of the observable. The physically relevant quantity is  $\sigma_{\text{pair}}(n) = \sigma_c(n) \cdot \sigma_{q-c}(n)$  on conjugate pairs  $\{c, q-c\}$ . The exponent doubling  $\delta_{\text{pair}} = 2\delta_c \approx 7.44$  resolves the gap (**structural**).

**O17 [17]:** Structural derivation.  $\rho_{q-c} = \bar{\rho}_c$  is forced by fibre-level admissibility; the pair structure is derived, not postulated (**proved**).

**O18 [25]:** Foundational derivation of the parity involution  $c \leftrightarrow q-c$ . BI parity is the only global symmetry of  $S_{\text{BI}}[\chi]$ , ensuring each fibre of  $\Pi$  is exactly  $\{\chi, -\chi\}$  (minimal fibre condition, **proved**, Theorem 4.1).

**O19 [2]:** The residual amplitude factor  $r(c, q) \in \{1, 2, 3, 4\}$  does not affect  $\delta_{\text{pair}}$ ; the observable is pipeline-independent (**proved**).

**O20 [36]:** Dynamical selection criterion for admissible modes via projective persistence (**structural**).

**O21 [44]:** Defines the observable saturation rank  $n_3^{\text{obs}}$  via  $\Sigma_c(n_3) = 3$  and constructs the transfer  $\delta_{\text{pair}} \rightarrow \beta^*$  (**structural**).

**O22 [43]:** BI projection locking forces  $n_3^{\text{obs}}$  onto a BFS shell (Theorem 3.2, **proved**).

**O23 [52]:** The threshold  $\Sigma_c(n_3) = 3$  follows from the quaternionic maximality of the admissible neutral generating set (**proved**).

**O24 [55]:** Capstone of the analytical O-series. The Verticality Lemma (“ker  $\Pi$  is free, Im  $\Pi$  is constrained”) closes the unconditional transfer chain  $c_\chi \rightarrow \delta_{\text{pair}} \rightarrow \beta^*$  (Corollary 4.2:  $\beta^* \approx 0.126$ , **proved**).

#### 4.6 O25–O29 and PTO: Numerical Validation and Sector Identification

**O25 [50]:** Systematic pair-level numerical campaign. Computes  $\delta_{\text{pair}}$  across all  $(q-1)/2$  conjugate pairs with  $M = 50$  independent block samples per pair, for  $q \in \{29, 61, 101, 151, 211\}$ . Results:  $\bar{\delta}_{\text{pair}} \in \{8.89, 9.98, 9.55, 9.13, 8.78\}$  with inter-pair standard deviation decreasing from 0.54 to 0.11, confirming  $\delta_{\text{pair}}$  as a structural invariant. After the O14 normalisation correction,

corrected values  $\delta_{\text{corr}}(q) \in [7.7, 9.0]$  fall consistently within the admissible window  $[7.4, 10.6]$  for  $q \in \{61, 101, 151, 211\}$ . Direct extrapolation to  $q \rightarrow \infty$  is shown to be structurally degenerate; the central open direction is the analytical control of  $n_1(q)/q$  (**structural**).

**O26** [39]: Dictionary between conjugate Weil blocks and rank-1 matrices in  $M_2(\mathbb{C})$ , connecting the Heisenberg fibre to  $2I \subset \text{SU}(2)$  (**structural**).

**O27** [41]: Every admissible morphism  $\Phi_{q,\rho}$  satisfying involution equivariance, quadratic compatibility, and naturality factors through  $\mathfrak{su}(2)$  (**proved**).

**O28** [5]: Two measurements from the Q5a-O5 checkpoints. Part A: the BFS window ratio  $n_1(q)/q$  decreases from 0.138 to 0.066 over  $q \in \{29, 61, 101, 151, 211\}$  (Scenario S2, pre-asymptotic regime); OLS fit  $n_1(q) = \hat{\alpha}q + \hat{\beta}$  gives  $\hat{\alpha} \approx 0.053$ ;  $\delta_{\text{corr}}(q) \in [7.4, 10.6]$  for all tested primes, confirming O25. Part B: formal effective-dimension computation of O26 Criterion 5.4 in  $\text{End}(H_{\text{eff}})$ ,  $H_{\text{eff}} = \mathbb{C}^3$  (HEFF\_DIM = 3, consistent with  $\Sigma_c(n_3) = 3$  from O23). The covariance  $\mathcal{C}_c$  has rank exactly 3 for every conjugate pair and every  $q \in \{61, 101, 151\}$  (100% of pairs), with invariant eigenvalue structure  $[\lambda_1 : \lambda_2 : \lambda_3] = [1 : \frac{1}{2} : \frac{1}{2}]$ . The exact degeneracy  $\lambda_2 = \lambda_3$  is a structural consequence of the Born–Infeld parity involution  $c \leftrightarrow q - c$  (O18), which correlates conjugate projections and constrains  $\mathcal{C}_c$  to a symmetric subspace. The gap from  $d_\rho^2 = 4$  reflects  $H_{\text{eff}} \neq V_\rho$ ; analysis of  $V_\rho \subset H_{\text{eff}}$  is carried out in O29.

**O29** [48]: Spin- $\frac{1}{2}$  sector identification via the symmetric rank formula (**structural**, numerically confirmed). The Born–Infeld parity constraint  $\pi_{q-c}(v) = \overline{\pi_c(v)}$  (O18) is proved to force every outer product  $M_j = \pi_c(v_j) \otimes \pi_{q-c}(v_j)^*$  into the complex symmetric subspace  $\text{Sym}(V_\rho, \mathbb{C}) \subset \text{End}(H_{\text{eff}})$ . This is an algebraic identity, not lifted by any complex-linear reparametrisation. The resulting covariance rank satisfies the *symmetric rank formula*  $r_{\text{eff}} = d_\rho(d_\rho + 1)/2$ , whose unique positive integer inversion at  $r_{\text{eff}} = 3$  gives  $d_\rho = 2$ , confirming the spin- $\frac{1}{2}$  sector. Numerically:  $r_{\text{eff}} = 3$  universally for  $q = 61$  (30/30 pairs) and  $q = 151$  (75/75 pairs) with zero inter-pair variance; at  $q = 101$ , modal  $r_{\text{eff}} = 3$  is confirmed by a dedicated multi-sample analysis ( $M = 5$  resamples per anomalous pair, 89% of cells satisfy  $r_{\text{eff}} = 3$ ), resolving the single-sample anomaly as a Monte Carlo artefact. The O26 Criterion 5.4 is reformulated for conjugate-pair data:  $d_\rho^2 = 4 \rightarrow d_\rho(d_\rho + 1)/2$ ; spin- $\frac{1}{2}$  satisfies the reformulated criterion exactly.

**PTO** [38]: The cumulative projected capacity  $I(n) = \sigma_{\text{pair}}(0) - \sigma_{\text{pair}}(n)$  is strictly non-decreasing and hence compatible with an arrow of time. Confirmed numerically for  $q \in \{29, 61, 101, 151\}$  (**structural**).

## 5 Branch III: Q-Series and Physical Applications

### 5.1 Q-Series: Quantum Mechanics and Spacetime Emergence

The Q-series derives quantum-mechanical and geometric structure from the Foundation axioms, taking as input the fibre identification  $F_n \cong V_\rho$  and selected O-series results.

**Q1** [18]: Phase coherence is a structural consequence of projective admissibility (not a postulate). Theorem 2.7 proves that any admissible transition preserves the BI indiscernibility of conjugate Weil blocks  $\rho_c$  and  $\rho_{q-c}$ , maintaining the metaplectic phase coherence of the fibre. The singlet correlator  $E(\hat{a}, \hat{b}) = -\hat{a} \cdot \hat{b}$  and the Tsirelson bound are derived from admissibility, O18, and O23 (**proved**; numerically validated for  $q = 29$ ).

**Q2** [40]: Extends the programme to spin- $\frac{3}{2}$ . On the binary icosahedral group  $2I$ , the spin- $\frac{1}{2}$  and spin- $\frac{3}{2}$  sectors share the Laplacian eigenvalue  $\lambda_{1/2} = \lambda_{3/2} = 18$  (co-admissibility). The Born rule, singlet correlator, and Tsirelson bound are derived for spin- $\frac{3}{2}$  without new postulates.  $\text{SU}(2)$  is established as the stable fixed point of the admissibility flow (**proved** for spin- $\frac{3}{2}$ ; general- $j$  extension **open**).

**Q5a** [22]: Under Mosco hypotheses, the admissibility forms  $\mathcal{E}_q$  on  $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$  converge to  $L_\Pi = -A\partial_x^2$  on  $L^2(\mathbb{R})$  (**structural**, conditional).

**Q5b** [9]: Under the lifting hypothesis [H-lift], the BFS shell stratification of  $\text{Heis}_3(\mathbb{R})$  produces a four-dimensional Lorentzian spacetime, with the temporal direction identified with the BFS



depth (**structural**, conditional).

**Q6a** [19]: Gauge charges are the invariants of the projection-fibre equivalence classes.  $U(1)$  and  $SU(2)$  are identified as fixed points of the admissibility flow.  $SU(3) \times SU(2) \times U(1)$  is proposed as a composite (**structural**;  $SU(3)$ : **open**).

**Q6b** [14]: Effective Lorentzian geometry from the admissibility filter  $\Pi_q$ ; inherits Q5a–Q5b hypotheses (**structural**, conditional).

## 5.2 Gravity and Spacetime Track

**Spectral2** [42]: Derives operational metric distances from graph Laplacians on a relational substrate without any background manifold (**structural**).

**Gravity** [21]: Derives  $G_{\mu\nu} = 8\pi G_N T_{\mu\nu}^{(\Pi)}$  from the metric variation of  $S_\Pi[g] = \frac{1}{2} \log \det' A_g$ , via heat-kernel and zeta regularisation (**proved** at leading IR order).

**Thermodynamics** [23]: Develops a local thermodynamic interpretation. The diagonal heat kernel yields a local multiplier field  $\beta(x)^{-1} = \beta_0^{-1} - \frac{1}{6}R(x) + O(\nabla^2 R)$ , and a spectral first law  $\delta S_\Pi = \int \beta(x)^{-1} \delta E_\Pi(x)$  recasts the Einstein equation as a spectral equilibrium condition (**structural**; Lorentzian extension and matter coupling: **open**).

**Lorentz** [11]: Lorentzian completion of the spectral action. Causal propagation and gravitational wave propagation follow from the hyperbolic continuation (**proved** for the Lorentzian extension; GW spectrum: **structural**).

## 5.3 Standard Model Companion Papers

**BornInfeld** [10]: BI action as the unique minimal local representation consistent with a universal bound on admissible relational gradients (**proved**).

**TopologicalInvariants** [53]: Charge quantisation ( $\pi_1(S^1) \cong \mathbb{Z}$ ) and absence of magnetic monopoles from the Hopf fibration  $S^1 \hookrightarrow S^3 \rightarrow S^2$  of the projection fibre (**proved**;  $\pi_2(\mathcal{C}_{\text{eff}}) = 0$  beyond the canonical model: **open**, Remark 4.2).

**SMchargemass** [12]: Mass and electric charge as BI-saturated responses (**structural**; baryon-to-photon ratio  $\eta$ : **open**).

**Bell** [8]: Bell-inequality violations from non-injectivity alone, without nonlocality or additional hidden variables (**proved**).

**SchwingerLamb** [20]: Fine-structure constant  $\alpha$  as a projective invariant; Lamb shift and Schwinger correction rederived in the pre-geometric framework (**structural**; quantitative radiative corrections: **heuristic**).

## 5.4 Cosmological and Condensed-Matter Applications

**Cosmology** [45]: Flat galaxy rotation curves and the Hubble tension from a common effective mechanism; no dark matter or modified gravity (**structural**; quantitative fit: **heuristic**).

**Superconductivity** [54]: Projective phase locking as the structural origin of superconducting coherence across BCS, cuprate, and nickelate systems (**structural**).

# 6 Dependency Structure

Figure 1 presents the logical dependency DAG. Three entry paths emerge for external readers.

**Foundational path.**  $\text{ENI} \rightarrow \text{Foundation} \rightarrow \text{HeisenbergStructure} \rightarrow \text{noscale}$ . Requires least prior knowledge; establishes the axiomatic core.

**Spectral path.**  $\text{BornInfeld} \rightarrow \text{SpAdm} \rightarrow \text{SpRel} \rightarrow \text{O1} \rightarrow \dots \rightarrow \text{O24} \rightarrow \text{O25} \rightarrow \text{O28} \rightarrow \text{O29}$ . Develops the spectral machinery and the transfer chain.



**Physical path.** Foundation + O18 + O22 + O23  $\rightarrow$  Q1  $\rightarrow$  Q2  $\rightarrow$  Q5a  $\rightarrow$  Q5b  $\rightarrow$  Q6a + Q6b;  
parallel: Spectral2  $\rightarrow$  Gravity  $\rightarrow$  Thermodynamics  $\rightarrow$  Lorentz.

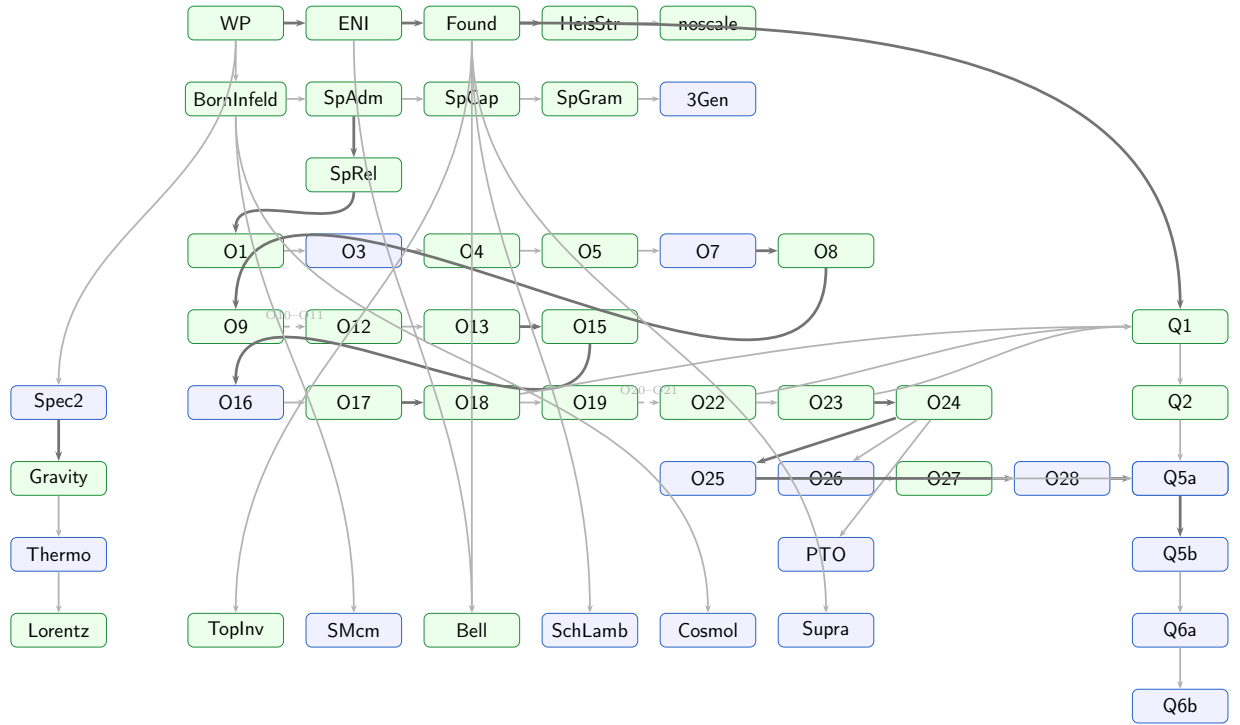


Figure 1: Logical dependency DAG of the Cosmochrony programme. Colour coding: ■ **proved**, ■ **structural**, ■ **heuristic**, ■ **open**. Thick arrows indicate the main logical spine; thin arrows indicate supporting dependencies. Dashed arrows with labels abbreviate linear sub-chains. The interactive version with clickable nodes is provided as a companion HTML file.

## 7 Full Paper Inventory

Table ?? lists all papers in the programme. The **Key** column gives the cosmochnrony-bibliography citation key.

| Label      | Short title                      | Key  | Main contribution                                                                                                                                           | Status     |
|------------|----------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| WP         | White Paper                      | [13] | Programme overview; substrate $\chi$ ; iterated projection; BI constraint; relaxation cascade                                                               | structural |
| ENI        | Emergence & non-injectivity      | [28] | No-go: genuine emergence $\Leftrightarrow \Pi$ non-injective; projection entropy $S_\Pi$                                                                    | proved     |
| Found      | Foundation                       | [4]  | Axioms A1–A4; derives $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ and $V_\rho$ as theorems (Thm 5.6)                                                            | proved     |
| HeisStr    | Heisenberg structure             | [27] | Detailed proof of Found. Thm 5.6; uncertainty principle from A3 alone                                                                                       | proved     |
| noscale    | No external scale                | [26] | No independent dimensional parameter beyond $c_\chi$                                                                                                        | proved     |
| BornInfeld | Born–Infeld geometry             | [10] | BI action as unique minimal bounded-flux representation                                                                                                     | proved     |
| SpAdm      | Spectral admissibility ( $Q_8$ ) | [46] | $A_n^{\max} = c_\chi/\sqrt{\lambda_n}$ ; first tractable instance                                                                                           | proved     |
| SpCap      | Spectral capacity                | [47] | Capacity functional $\mathcal{C}(G, S)$ ; binary-polyhedral maximality proved via adjoint identity $\hat{A}_{\text{adj}} = 2M - I$ and icosahedral isotropy | proved     |
| SpGram     | Gram rigidity (v1.0.1)           | [31] | Neutral-generator geometry from $G_{st} = -\frac{1}{2}\chi_\rho(st)$                                                                                        | proved     |
| 3Gen       | Three generations                | [51] | Three-generation structure from ADE stratigraphic levels                                                                                                    | structural |
| SpRel      | Spectral relaxation              | [6]  | $\Lambda_{\text{proj}} \asymp \lambda_2$ for LPS expander families                                                                                          | proved     |
| O1         | Variable-valence graphs          | [37] | LPS saturation is a fixed-valence artefact; growing valence restores window                                                                                 | proved     |
| O3         | Mass hierarchy                   | [35] | First SM connection: $\beta^* \in (0.09, 0.13)$ from lepton mass ratios                                                                                     | structural |
| O4         | Upper bound $\beta \leq 1$       | [49] | BI flux + Cheeger $\Rightarrow \beta \leq 1$                                                                                                                | proved     |
| O5         | Rep-theoretic obstruction        | [3]  | Fixed-repr. approaches cannot close the $\beta$ gap                                                                                                         | proved     |
| O6         | Matrix redundancy                | [24] | Matrix-level confinement of cascade exponent                                                                                                                | proved     |
| O7         | Continuum target                 | [33] | Target: $\delta \in [7.4, 10.6]$ on 3-step fingerprints                                                                                                     | structural |
| O8         | LPS geometric obstruction        | [1]  | Exponential shell growth $\Rightarrow$ usable window = $O(\log q)$ BFS steps                                                                                | proved     |
| O9         | Heisenberg graphs                | [34] | Structural transition: $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ with $ B_n  \sim n^4$ replaces LPS                                                           | proved     |
| O10        | Large- $q$ computation           | [32] | Confirms $n^4$ growth; TensorSketch obstruction identified                                                                                                  | proved     |

*continued. . .*

| Label | Short title                           | Key  | Main contribution                                                                                                                                                                                                                                      | Status     |
|-------|---------------------------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| O11   | Fourier-character proxy               | [56] | Bidimensional proxy overcomes TensorSketch obstruction                                                                                                                                                                                                 | structural |
| O12   | Exact Weil blocks                     | [15] | $\hat{\delta}_{\text{exact}} \approx 4.3\text{--}4.8$ for $q \leq 211$ ; final obstruction resolved                                                                                                                                                    | proved     |
| O13   | Extended prime range                  | [7]  | $\hat{\delta}(q) : 4.80 \rightarrow 3.59$ ; finite-size hypothesis refuted                                                                                                                                                                             | proved     |
| O14   | Observable-class correction           | [30] | Corrects $\delta \mapsto \beta^*$ via block normalisation                                                                                                                                                                                              | structural |
| O15   | Block-level no-go                     | [29] | No reweighting of individual blocks reaches $\delta \in [7.4, 10.6]$ (Cor. 4.8)                                                                                                                                                                        | proved     |
| O16   | Pair observable                       | [16] | $\sigma_{\text{pair}} = \sigma_c \sigma_{q-c}$ ; exponent doubling $\delta_{\text{pair}} \approx 7.44$                                                                                                                                                 | structural |
| O17   | Structural derivation of pair         | [17] | $\rho_{q-c} = \bar{\rho}_c$ forced by fibre admissibility: pair is derived, not postulated                                                                                                                                                             | proved     |
| O18   | BI parity involution                  | [25] | Minimal fibre condition $\{\chi, -\chi\}$ ; BI parity is the unique global symmetry of $S_{\text{BI}}$                                                                                                                                                 | proved     |
| O19   | Canonical normalisation               | [2]  | $r(c, q)$ does not affect $\delta_{\text{pair}}$ ; observable pipeline-independent                                                                                                                                                                     | proved     |
| O20   | Persistence criterion                 | [36] | Dynamical selection of admissible modes via projective persistence                                                                                                                                                                                     | structural |
| O21   | Shell saturation                      | [44] | Observable rank $n_3^{\text{obs}}$ via $\Sigma_c(n_3) = 3$ ; transfer $\delta_{\text{pair}} \rightarrow \beta^*$                                                                                                                                       | structural |
| O22   | Projection locking                    | [43] | BI projection locking forces $n_3^{\text{obs}}$ onto a BFS shell (Thm 3.2)                                                                                                                                                                             | proved     |
| O23   | Quaternionic minimality               | [52] | $\Sigma_c(n_3) = 3$ from quaternionic maximality of neutral generators                                                                                                                                                                                 | proved     |
| O24   | Observable rank rigidity              | [55] | Verticality Lemma; unconditional transfer $c_\chi \rightarrow \delta_{\text{pair}} \rightarrow \beta^* \approx 0.126$ (Cor. 4.2)                                                                                                                       | proved     |
| O25   | Numerical campaign                    | [50] | $\bar{\delta}_{\text{pair}} \in \{8.89, 9.98, 9.55, 9.13, 8.78\}$ for $q \in \{29, 61, 101, 151, 211\}$ ; inter-pair $\sigma_q : 0.54 \rightarrow 0.11$ ; $\delta_{\text{corr}} \in [7.7, 9.0] \subset [7.4, 10.6]$                                    | structural |
| O26   | Quadratic completion                  | [39] | Dictionary: Weil blocks $\leftrightarrow$ rank-1 matrices in $M_2(\mathbb{C})$ ; connects $\text{Heis}_3$ to $2I \subset \text{SU}(2)$                                                                                                                 | structural |
| O27   | Quaternionic rigidity                 | [41] | Every admissible $\Phi_{q,\rho}$ satisfying three structural constraints factors through $\mathfrak{su}(2)$                                                                                                                                            | proved     |
| O28   | Window calibration & $r_{\text{eff}}$ | [5]  | $n_1(q)/q$ decreasing (Scenario S2); $\delta_{\text{corr}} \in [7.4, 10.6]$ confirmed; $r_{\text{eff}} = 3$ in $\text{End}(H_{\text{eff}})$ , eigenstructure $[1 : \frac{1}{2} : \frac{1}{2}]$ invariant across all pairs and $q \in \{61, 101, 151\}$ | structural |

*continued...*

| Label   | Short title                        | Key  | Main contribution                                                                                                                                                                                                                                                                      | Status     |
|---------|------------------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| O29     | Symmetric rank formula & sector ID | [48] | BI parity forces $M_j \in \text{Sym}(V_\rho, \mathbb{C})$ ; $r_{\text{eff}} = d_\rho(d_\rho + 1)/2$ ; unique inversion at $r_{\text{eff}} = 3$ gives $d_\rho = 2$ (spin- $\frac{1}{2}$ ); confirmed for $q \in \{61, 101, 151\}$ ; $q = 101$ anomaly resolved by multi-sample analysis | structural |
| PTO     | Temporal ordering                  | [38] | $I(n) = \sigma_{\text{pair}}(0) - \sigma_{\text{pair}}(n)$ is a non-decreasing temporal proxy; confirmed for $q \in \{29, 61, 101, 151\}$                                                                                                                                              | structural |
| Q1      | Phase coherence                    | [18] | Phase coherence from admissibility (Thm 2.7); singlet correlator and Tsirelson bound derived                                                                                                                                                                                           | proved     |
| Q2      | Beyond spin- $\frac{1}{2}$         | [40] | Spin- $\frac{3}{2}$ quantum structure; $\text{SU}(2)$ as admissibility fixed point                                                                                                                                                                                                     | proved     |
| Q5a     | Large- $q$ operator limit          | [22] | Mosco convergence $\mathcal{E}_q \rightarrow L_\Pi = -A\partial_x^2$ ; conditional                                                                                                                                                                                                     | structural |
| Q5b     | 4D Lorentzian geometry             | [9]  | BFS stratification of $\text{Heis}_3(\mathbb{R}) \rightarrow 4\text{D}$ Lorentzian spacetime; conditional                                                                                                                                                                              | structural |
| Q6a     | Gauge structure                    | [19] | $\text{U}(1)$ and $\text{SU}(2)$ from fibre fixed points; $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ proposed                                                                                                                                                               | structural |
| Q6b     | Effective spacetime                | [14] | Effective Lorentzian geometry from $\Pi_q$ ; inherits Q5a–Q5b hypotheses                                                                                                                                                                                                               | structural |
| Spec2   | Relational space-time              | [42] | Operational metric from graph Laplacians; spacetime from spectral correlations                                                                                                                                                                                                         | structural |
| Gravity | IR Einstein response               | [21] | $G_{\mu\nu} = 8\pi G_N T_{\mu\nu}^{(\Pi)}$ from $\delta S_\Pi/\delta g$ at leading IR order                                                                                                                                                                                            | proved     |
| Thermo  | Spectral thermodynamics            | [23] | Local spectral first law; Einstein equation as spectral equilibrium condition                                                                                                                                                                                                          | structural |
| Lorentz | Gravitational waves                | [11] | Lorentzian completion; causal propagation; GW spectrum                                                                                                                                                                                                                                 | proved     |
| TopInv  | Topological invariants             | [53] | Charge quantisation ( $\pi_1 \cong \mathbb{Z}$ ), no monopoles, chirality (Hopf fibration)                                                                                                                                                                                             | proved     |
| SMcm    | Charge, mass, inertia              | [12] | Mass and charge as BI-saturated responses                                                                                                                                                                                                                                              | structural |
| Bell    | Bell non-locality                  | [8]  | Bell violations from non-injectivity alone                                                                                                                                                                                                                                             | proved     |
| SchLamb | Fine-structure constant            | [20] | $\alpha$ as projective invariant; Lamb shift and Schwinger correction                                                                                                                                                                                                                  | structural |
| Cosmol  | Rotation curves / Hubble           | [45] | Flat rotation curves and Hubble tension from a common effective mechanism                                                                                                                                                                                                              | structural |
| Supra   | Superconductivity                  | [54] | Projective phase locking as origin of superconducting coherence                                                                                                                                                                                                                        | structural |

## 8 Open Problems and Roadmap

### 8.1 Branch II (O-Series)

**Spin- $\frac{1}{2}$  sector identification (O29, closed).** O29 proves that the Born–Infeld parity constraint (O18) forces every outer product  $M_j = \pi_c(v_j) \otimes \pi_{q-c}(v_j)^*$  into the complex symmetric subspace  $\text{Sym}(V_\rho, \mathbb{C})$ , an algebraic identity not lifted by any complex-linear reparametrisation. The symmetric rank formula  $r_{\text{eff}} = d_\rho(d_\rho + 1)/2$ , combined with the observed  $r_{\text{eff}} = 3$ , uniquely identifies  $d_\rho = 2$  (spin- $\frac{1}{2}$ ). This result is confirmed numerically for  $q \in \{61, 101, 151\}$  with zero inter-pair variance at  $q = 61$  and  $q = 151$ ; the  $q = 101$  anomaly is resolved as a single-sample Monte Carlo artefact by a dedicated multi-sample analysis. The O26 Criterion 5.4 is reformulated for conjugate-pair data ( $d_\rho^2 \rightarrow d_\rho(d_\rho + 1)/2$ ) and satisfied exactly. The remaining open directions within Branch II are: (a) analytical derivation of the eigenvalue ratio  $\lambda_2/\lambda_1$  from the group structure; (b) extension of the computation to  $q \in \{211, 307, 401\}$ ; (c) a measurement protocol using non-conjugate block data to probe the full rank  $d_\rho^2 = 4$ .

**Analytical control of  $n_1(q)/q$ .** The BFS window ratio  $n_1(q)/q$  continues to decrease through  $q = 211$  (Scenario S2, O28); the asymptotic constant  $\hat{\alpha}$  is bounded below by 0.053 but not yet stabilised. Extending the campaign to  $q \in \{307, 401\}$  would constrain  $\hat{\alpha}$  further.

**Notation consolidation:**  $c_\chi \rightarrow c_{\text{BI}}$ . A coordinated rename of the BI saturation constant across all papers is planned and deferred pending consistency with already-deposited O-series papers.

### 8.2 Branch III (Q-Series and Applications)

**Unconditional continuum limit (Q5a).** Lifting the Mosco and lifting hypotheses to unconditional results is the main open problem in the geometric emergence track.

**$\pi_2(\mathcal{C}_{\text{eff}}) = 0$  beyond the canonical model (TopologicalInvariants, Remark 4.2).** General proof of this topological property is open.

**Full SM gauge group (Q6a).** A structural derivation of  $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$  as a composite of admissibility fixed points is proposed but not yet proved.

**Baryon-to-photon ratio (SMchargemass).** Quantitative derivation of  $\eta \sim 10^{-10}$  is open.

**$V - A$  chiral structure.** Formal derivation of the  $V - A$  coupling from spectral selection is open.

**Spin- $j$  extension (Q2).** The structural derivation of quantum mechanics for all  $j$  in the  $Q_8 \hookrightarrow 2I \hookrightarrow \text{SU}(2)$  chain is open beyond spin- $\frac{3}{2}$ .

**Thermodynamics: Lorentzian extension and matter coupling.** The local spectral first law is derived in the Riemannian setting. Analytic continuation and the explicit form of  $W_\Pi$  for matter fields are open.

## 9 Conclusion

The Cosmochrony programme has established a coherent, multi-level derivation of physical structure from a single primitive. Branch I (Foundation, ENI, HeisenbergStructure [27]) is formally closed: the Weil representation of  $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$  is a theorem. The binary-polyhedral maximality conjecture (SpectralCapacity [47]) is now proved:  $2I$  uniquely maximises  $\mathcal{C}_\alpha(G, S)$  for all  $\alpha > 0$



among finite  $SU(2)$  subgroups with neutral generating sets, closing the precursor programme for the O-series. Branch II (O-series) has closed the unconditional transfer chain  $c_\chi \rightarrow \delta_{\text{pair}} \rightarrow \beta^*$  analytically (O24) and numerically validated for  $q \in \{29, 61, 101, 151, 211\}$  (O25). The spin- $\frac{1}{2}$  sector identification is now complete (O29): the symmetric rank formula  $r_{\text{eff}} = d_\rho(d_\rho + 1)/2$  uniquely identifies  $d_\rho = 2$  from the observed  $r_{\text{eff}} = 3$ , confirmed numerically for  $q \in \{61, 101, 151\}$ . Branch III (Q-series and companions) derives quantum mechanics, spacetime, and SM observables at the structural level, with several results conditional on explicitly stated hypotheses.

For an external reader, three entry sequences are suggested. For the quantum structure thread:  $\text{ENI} \rightarrow \text{Foundation} \rightarrow \text{O18} \rightarrow \text{O22} \rightarrow \text{O23} \rightarrow \text{Q1}$ . For the spectral mass hierarchy thread:  $\text{ENI} \rightarrow \text{Foundation} \rightarrow \text{O16} \rightarrow \text{O24}$ . For the spacetime thread:  $\text{Spectral2} \rightarrow \text{Gravity} \rightarrow (\text{Thermodynamics or Lorentz})$ .

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